system, then followed by the cardiovascular system. In paediatric patients, when a local anaesthetic is administered under general anaesthesia, it is difficult to detect the first signs of toxicity to the local anaesthetic.

Central nervous system toxicity

Initially, symptoms include agitation, a feeling of intoxication, a sensation of numbness in the lips and tongue, paresthesias around the mouth, dizziness, vision and hearing disturbances, and buzzing in the ears. Speech problems, muscle stiffness and spasms are more serious symptoms and precede generalised seizures. Respiratory arrest may even occur in severe cases. Acidosis increases the toxic effects of local anaesthetics. Recovery depends on the metabolism of the local anaesthetic and the distribution away from the central nervous system. This occurs quickly providing large amounts of the drug are not injected.

Cardiovascular system

The symptoms associated to the local anaesthetic may include blood pressure drop, bradycardia, arrhythmia, and cardiac arrest as a result of high systemic concentrations of local anaesthetic.

The symptoms associated to epinephrine are heat sensation, sweating, heart rhythm acceleration, headaches, blood pressure increase, anginous disorders, tachycardia, tachyarrhythmia, and cardiac arrest.

Treatment

General basic measures

If adverse reactions arise the application of the local anaesthetic has to be stopped. Measures should focus on maintenance/restoration of the vital functions of respiration and circulation, oxygen administration and intravenous access.

Special measures

- Hypertension: Elevation of the upper body, if necessary sublingual nifedipine.
- Convulsions: Protect patients from concomitant injuries, if necessary benzodiazepines (e.g. diazepam iv).
- Hypotension: Horizontal position, raise legs up, if necessary intravascular infusion of a complete electrolyte solution iv, vasopressors (e.g. ethylephrine iv).
- Bradycardia: Atropine iv.
- Anaphylactic shock: Infusion of a complete electrolyte solution, if necessary epinephrine iv, cortisone iv; contact emergency physician.
- Cardiovascular arrest: Immediate cardiopulmonary resuscitation, contact emergency physician.

INCOMPATIBILITIES

In solutions with epinephrine, the mixture with alkaline solutions may cause a rapid degradation of the vasoconstrictor agent, as well as a greater risk of precipitation.

STORAGE INFORMATION

Special precautions for storage

Store below 30°C and protected from light.

Shelf life

18 months.

Nature and contents of container

Neutral, hydrolytic class type I glass cartridges, closed on one side with a grey-coloured bromobutyl stopper and on the other side with a bromobutyl coated disc and aluminium cap.

XILONIBSA® 2% is packed in boxes containing 50 cartridges.

Special precautions for disposal

Cartridges for single use. Cartridges should not be used with other patients. The remaining of the product should be discarded.

Any unused product or waste material should be disposed of in accordance with local requirements.

DATE OF REVISION OF THE TEXT

November, 2023

MANUFACTURER

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PRODUCT REGISTRATION HOLDER

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Selangor, Malaysia
Phone No. +603 5525 2581

20002429 OF

XILONIBSA® 2%

Solution for injection

(Local injection / Oromucosal use)

COMPOSITION

1 ml solution for injection contains:

Lidocaine hydrochloride 20.0 mg
(equivalent to Lidocaine hydrochloride monohydrate 21.33 mg)
Epinephrine 0.0125 mg (equivalent to Epinephrine tartrate 0.02274mg)

Excipients:

Sodium chloride	4.57	mg
Sodium metabisulphite	0.50	mg
Citric acid monohydrate	0.2	mg

Hydrochloric acid, Sodium hydroxide, Water for injection.

1 cartridge (1.8ml) contains 36mg of lidocaine hydrochloride and 0.0225 mg of epinephrine (equivalent to 0.04095mg of epinephrine tartrate).

PRODUCT DESCRIPTION

Solution for injection, clear and colourless.

THERAPEUTIC INDICATIONS

Infiltration and nerve-block dental local anaesthesia.

POSOLOGY AND METHOD OF ADMINISTRATION

Posology

The dosage should be adjusted individually according to the area to be anaesthetised, vascularisation of the tissues and anaesthetic technique to be used.

For infiltrations or terminal anaesthesia, 1 ml of XILONIBSA® 2% suffices in most cases. In the case of nerve-block anaesthesia, the dose will be 1.5-2 ml.

Maximum dose

Adults:

The maximum dose over a period of 24 hours is 500 mg of lidocaine (for a person of 70 kg), which must not exceed 7 mg/kg of body weight in adults.

Children:

The dose will be adjusted according to the patient's age and weight, as well as the type of surgery to be performed, not exceeding 5 mg/kg of body weight. The use of XILONIBSA® 2% is contraindicated in children under 4 years of age.

Special population:

Increased plasma levels of XILONIBSA® 2% can occur in older patients due to diminished metabolic processes and lower distribution volume.

The risk of accumulation of XILONIBSA® 2% is increased in particular after repeated application. A similar effect can ensue from severely impaired hepatic function (see section *Special warnings and precautions for use*), thus a lower dose is recommended in these cases.

Method of administration

For injection/oromucosal use.

For use in dental anaesthesia only.

To avoid intravascular injection, aspiration control at least in two planes (rotation of the needle by 180°) must always be carefully undertaken. The injection rate should not exceed 0.5 ml in 15 seconds, i.e. 1 cartridge per minute.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic properties

Pharmacotherapeutic group: local anaesthetics: amides, ATC code: N01BB52.

As other local anaesthetics, lidocaine reversibly blocks the propagation of the impulse along the nerve fibers, thus preventing the mobility of sodium ions through the nerve membrane.

At high doses lidocaine has a quinidine-like action on the myocardium, i.e., cardiac depressant. All local anaesthetics stimulate the central nervous system and may produce anxiety, restlessness and tremors.

The onset and duration of lidocaine action are increased by adding epinephrine as vasoconstrictor. Thus, the absorption of the anaesthetic is delayed and a greater concentration is obtained for a longer period. Also, the possibility of systemic adverse effects is reduced.

Pharmacokinetic properties

Lidocaine is rapidly absorbed after intramuscular injection and oromucosal injection. The distribution volume (Vdis) is 1.30 L/kg and the clearance (Cl) is 0.85 L/kg/hr. Lidocaine undergoes first-pass metabolism in the liver and less than 10% of the dose is excreted unchanged via the kidneys. Elimination half-life ($t_{1/2}$) is 1.6 hours.

Preclinical safety data

Non-clinical safety data reveal no special hazard for humans based on conventional studies of safety pharmacology, repeated dose toxicity, genotoxicity, carcinogenic potential and toxicity to reproduction.

As for other amide-type local anaesthetics, the active substance in high doses may cause reactions on the central nervous system and the cardiovascular system (see section *Undesirable effects*).

CONTRAINDICATIONS

XILONIBSA® 2% is contraindicated in case of hypersensitivity to the active ingredients or any other of the components. The use of XILONIBSA® 2% is contraindicated in children younger than 4 years of age.

Due to the content in lidocaine, XILONIBSA® 2% is contraindicated in case of:
- known allergy or hypersensitivity to local anaesthetics of the amide type
- severe disorders of atrioventricular conduction not compensated by a pacemaker.
- deficiency in plasma cholinesterase activity
- severe blood coagulation dysfunction
- degenerative nervous disorders

Due to the content in epinephrine (adrenaline); XILONIBSA® 2% is contraindicated in case of:
- Unstable angina pectoris
- Recent myocardial infarction
- Recent coronary artery bypass surgery
- Refractory arrhythmias and paroxistic or high rate tachycardia, continuous arrhythmia
- Severe untreated or uncontrolled hypertension
- Untreated or uncontrolled congestive heart failure
- Concomitant treatment with monoamine oxidase (MAO) inhibitors or tricyclic antidepressants

Due to the content in metabisulphite, XILONIBSA® 2% is contraindicated in case of:
- Allergy or hypersensitivity to sulphite
- Severe bronchial asthma

SPECIAL WARNINGS AND PRECAUTIONS FOR USE

Warnings

Inadvertent intravascular injection may be associated with convulsions, followed by central nervous system or cardiorespiratory arrest. Resuscitative equipment, oxygen and other resuscitative drugs should be available for immediate use.

It should be taken into consideration that during treatment with blood coagulation inhibitors (e.g. heparin or acetylsalicylic acid), an inadvertent vasopuncture when administering the local anaesthetic can lead to serious bleeding, and the hemorrhagic tendency may be increased (see section *Interaction With Other Medicinal Products and Other Forms of Interaction*).

Athletes should be warned that this medicinal product contains an active substance likely to yield a positive result in anti-doping tests.

The injection of this medicinal product must be avoided in infected area.

This preparation contains Sodium metabisulphite that may cause serious allergic type reactions in certain susceptible patients. Do not use if known to be hypersensitive to bisulphites.

This medicinal product contains less than 1 mmol sodium (23 mg) per 1 ml, i.e., essentially sodium-free.

Precautions for use

The product should be administered with caution in patients with impaired cardiovascular function since they may be less able to compensate for functional changes associated with the prolongation of atrioventricular conduction produced by these drugs (see section *Contraindications*).

XILONIBSA® 2% should be used with precaution in patients with:
- angina pectoris
- arteriosclerosis
- impaired blood coagulation
- diabetes mellitus
- severe hepatic impairment
- lung diseases – particularly allergic asthma
- epilepsy
- phaeochromocytoma
- narrow-angle glaucoma
- thyrotoxicosis

INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION

Due to its content in lidocaine, XILONIBSA® 2% should be used with caution in patients that receive concomitant medication similar in structure to local anaesthetics (for example, Ib-class antiarrhythmic drugs), since the toxic effects are additive.

Due to its content in epinephrine, XILONIBSA® 2% should be administered with caution in patients who are simultaneously receiving one of the following medications:
- Blood coagulation inhibitors (heparin), nonsteroidal anti-inflammatory drugs (NSAID), plasma substitutes (dextran), phenotiazines, butyrophenones: these drugs may reduce or reverse the vasopressor effect of epinephrine and may increase bleeding trend.
- Tricyclic antidepressants, monoamine oxidase inhibitors (MAOI), ergotamine-like oxytocic drugs, non-selective beta blockers like propanol: these drugs may increase the vasopressor effect of epinephrine and may lead to serious hypertension and bradycardia.

PREGNANCY AND LACTATION

Pregnancy

Even if there is no evidence from animal studies of harm to the foetus, XILONIBSA® 2% should not be given during pregnancy unless strictly necessary.

The administration of XILONIBSA® 2% during pregnancy may cause foetal bradycardia due to its content in local anaesthetic, as well as a decrease in intrauterine blood flow due to its content in epinephrine, especially in case of inadvertent intravascular injection.

Lactation

Lidocaine may enter the mother's milk, but in such small amounts that there is generally no risk of this affecting the neonate. It is not known whether adrenaline enters breast milk or not, but it is unlikely to affect the breast-fed child. Thus, it is considered that XILONIBSA® 2% may be used during lactation.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

The influence of XILONIBSA® 2% on the ability to drive or use machines is small or moderate, and may temporarily impair mental function and co-ordination depending on the dose of local anaesthetic.

The dentist has to assess in each case the possible impairment of safety when operating a motor vehicle or machinery. The patient should not leave the dental office earlier than at least 30 minutes after the injection.

UNDESIRABLE EFFECTS

Adverse effects strictly attributable to the local anaesthetic are limited. However, the physiological effects from nerve block are frequent, but these vary considerably depending on what type of block is administered.

Due to the content in lidocaine as a local anaesthetic, the following undesirable effects may occur:

Cardiovascular disorders

Rare (≥1/10,000, <1/1,000)

Drop in blood pressure, cardiac impulse conduction disorders, bradycardia, cardiovascular arrest.

Nervous system disorders

Rare (≥1/10,000, <1/1,000)

Metallic taste, tinnitus, dizziness, nausea, vomiting, anxiety, shaking, nervousness, nystagmus, headache, increase in respiratory rate. Paresthesias (loss of sensation, burning, tingling) of the lip, tongue, or both. Drowsiness, tonic-clonic seizures, coma and respiratory paralysis.

Respiratory disorders

Rare (≥1/10,000, <1/1,000)

Tachypnea, then bradypnea, which could lead to apnoea.

Allergic reactions

Very rare (< 1/10,000)

Rash, erythema, pruritus, oedema of the tongue, mouth, lips or throat and in more severe cases, anaphylactic shock.

Due to the content of epinephrine as a vasoconstrictor, the following undesirable effects can occur:

Cardiovascular disorders

Rare (≥1/10,000, <1/1,000)

Heat sensation, sweating, heart racing, migrainelike headache, blood pressure increase, angina pectoris disorders, tachycardias, tachyarrhythmias and cardiovascular arrest; acute oedematous thyroid swelling may not be discarded.

Due to the content of metabisulphite as excipient, the following undesirable effects can occur:

Allergic reactions

Very rare (<1/10,000)

Allergic reactions may particularly occur in bronchial asthmatics, which are manifested as vomiting, diarrhoea, wheezing, acute asthma attack, clouding of consciousness or shock.

OVERDOSE

Toxicity

Accidental intravascular injections of local anaesthetics may cause immediate systemic toxic reactions. In the event of overdose, systemic toxicity appears later (15–60 minutes after injection). Toxicity is manifested first in the central nervous system, then followed by the cardiovascular system. In paediatric patients, when a local anaesthetic is administered under general anaesthesia, it is difficult to detect the first signs of toxicity to the local anaesthetic.

Central nervous system toxicity

Initially, symptoms include agitation, a feeling of intoxication, a sensation of numbness in the lips and tongue, paresthesias around the mouth, dizziness, vision and hearing disturbances, and buzzing in the ears. Speech problems, muscle stiffness and spasms are more serious symptoms and precede generalised seizures. Respiratory arrest may even occur in severe cases. Acidosis increases the toxic effects of local anaesthetics. Recovery depends on the metabolism of the local anaesthetic and the distribution away from the central nervous system. This occurs quickly providing large amounts of the drug are not injected.

Cardiovascular system

The symptoms associated to the local anaesthetic may include blood pressure drop, bradycardia, arrhythmia, and cardiac arrest as a result of high systemic concentrations of local anaesthetic.

The symptoms associated to epinephrine are heat sensation, sweating, heart rhythm acceleration, headaches, blood pressure increase, anginous disorders, tachycardia, tachyarrhythmia, and cardiac arrest.

Treatment

General basic measures

If adverse reactions arise the application of the local anaesthetic has to be stopped. Measures should focus on maintenance/restoration of the vital functions of respiration and circulation, oxygen administration and intravenous access.

Special measures

- Hypertension: Elevation of the upper body, if necessary sublingual nifedipine.
- Convulsions: Protect patients from concomitant injuries, if necessary benzodiazepines (e.g. diazepam iv).
- Hypotension: Horizontal position, raise legs up, if necessary intravascular infusion of a complete electrolyte solution iv, vasopressors (e.g. ethylephrine iv).
- Bradycardia: Atropine iv.
- Anaphylactic shock: Infusion of a complete electrolyte solution, if necessary epinephrine iv, cortisone iv; contact emergency physician.
- Cardiovascular arrest: Immediate cardiopulmonary resuscitation, contact emergency physician.

INCOMPATIBILITIES

In solutions with epinephrine, the mixture with alkaline solutions may cause a rapid degradation of the vasoconstrictor agent, as well as a greater risk of precipitation.

STORAGE INFORMATION

Special precautions for storage

Store below 30°C and protected from light.

Shelf life

18 months.

Nature and contents of container

Neutral, hydrolytic class type I glass cartridges, closed on one side with a grey-coloured bromobutyl stopper and on the other side with a bromobutyl coated disc and aluminium cap.

XILONIBSA® 2% is packed in boxes containing 50 cartridges.

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DATE OF REVISION OF THE TEXT

November, 2023

MANUFACTURER

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PRODUCT REGISTRATION HOLDER

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Selangor, Malaysia
Phone No. +603 5525 2581

20002429 OF

XILONIBSA® 2%

Solution for injection

(Local injection / Oromucosal use)

inibsa

COMPOSITION

1 ml solution for injection contains:

Lidocaine hydrochloride 20.0 mg

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Epinephrine 0.0125 mg (equivalent to Epinephrine tartrate 0.02274mg)

Excipients:

Sodium chloride	4.57	mg
Sodium metabisulphite	0.50	mg
Citric acid monohydrate	0.2	mg

Hydrochloric acid, Sodium hydroxide, Water for injection.

1 cartridge (1.8ml) contains 36mg of lidocaine hydrochloride and 0.0225 mg of epinephrine (equivalent to 0.04095mg of epinephrine tartrate).

PRODUCT DESCRIPTION

Solution for injection, clear and colourless.

THERAPEUTIC INDICATIONS

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POSOLOGY AND METHOD OF ADMINISTRATION

Posology

The dosage should be adjusted individually according to the area to be anaesthetised, vascularisation of the tissues and anaesthetic technique to be used.

For infiltrations or terminal anaesthesia, 1 ml of XILONIBSA® 2% suffices in most cases. In the case of nerve-block anaesthesia, the dose will be 1.5-2 ml.

Maximum dose

Adults:

The maximum dose over a period of 24 hours is 500 mg of lidocaine (for a person of 70 kg), which must not exceed 7 mg/kg of body weight in adults.

Children:

The dose will be adjusted according to the patient's age and weight, as well as the type of surgery to be performed, not exceeding 5 mg/kg of body weight. The use of XILONIBSA® 2% is contraindicated in children under 4 years of age.

Special population:

Increased plasma levels of XILONIBSA® 2% can occur in older patients due to diminished metabolic processes and lower distribution volume.

The risk of accumulation of XILONIBSA® 2% is increased in particular after repeated application. A similar effect can ensue from severely impaired hepatic function (see section *Special warnings and precautions for use*), thus a lower dose is recommended in these cases.

Method of administration

For injection/oromucosal use.

For use in dental anaesthesia only.

To avoid intravascular injection, aspiration control at least in two planes (rotation of the needle by 180°) must always be carefully undertaken. The injection rate should not exceed 0.5 ml in 15 seconds, i.e. 1 cartridge per minute.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic properties

Pharmacotherapeutic group: local anaesthetics: amides, ATC code: N01BB52.

As other local anaesthetics, lidocaine reversibly blocks the propagation of the impulse along the nerve fibers, thus preventing the mobility of sodium ions through the nerve membrane.

At high doses lidocaine has a quinidine-like action on the myocardium, i.e., cardiac depressant. All local anaesthetics stimulate the central nervous system and may produce anxiety, restlessness and tremors.

The onset and duration of lidocaine action are increased by adding epinephrine as vasoconstrictor. Thus, the absorption of the anaesthetic is delayed and a greater concentration is obtained for a longer period. Also, the possibility of systemic adverse effects is reduced.

Pharmacokinetic properties

Lidocaine is rapidly absorbed after intramuscular injection and oromucosal injection. The distribution volume (Vdis) is 1.30 L/kg and the clearance (Cl) is 0.85 L/kg/hr. Lidocaine undergoes first-pass metabolism in the liver and less

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